MICROSOFT ACCESS

Library Management System

A beginner-friendly, click-by-click project guide

What you will build

A small relational database for storing books, library members, and book-borrowing records. The project includes tables, relationships, sample data, queries, data-entry forms, and a printable report.

Important: This guide is written for Microsoft Access on Windows. Take your own screenshots as you complete the steps; a screenshot checklist and suggested report order are included.

Project file: **LibraryManagement.accdb**

Student name: _____

Class / course: _____

Date: _____

Contents and project plan

Stage	What you will make	Suggested screenshot
1	Create and save the database	Access window showing the database name
2	Books, Members, and BookIssues tables	Each table in Design View
3	Relationships	Relationship diagram with both one-to-many links
4	Sample records	Datasheet View showing sample rows
5	Three queries	Query results in Datasheet View
6	Three forms	Each form in Form View with sample data
7	One report	Report Preview
8	Final testing	Navigation Pane and selected objects

Recommended workflow: complete one stage, save, take the screenshot listed for that stage, and only then move on. Keep screenshots in one folder so you can use them in your practical record or project report.

1. Create the database

1. Open Microsoft Access on your Windows computer.

2. On the start screen, choose **Blank database**.

3. In the File Name box, enter **LibraryManagement.accdb**.

4. Choose a folder you can find again, such as Documents > AccessProject.

5. Click **Create**. Access opens a new database and may create a default table named Table1.

If Access opens Table1: close it without saving if it is empty, or use the close button on its tab. We will create our own tables in Design View.

Screenshot 1: Capture the Access window after the database is created. Make sure the database name or Navigation Pane is visible.

2. Create the Books table

1. Choose **Create** on the ribbon, then click **Table Design**.

2. Enter the field names and data types exactly as shown below.

3. Click the row selector next to BookID, then click **Primary Key** on the ribbon. A key icon should appear next to BookID.

4. Press Ctrl+S and save the table as **Books**.

Field Name	Data Type	Notes
BookID	AutoNumber	Primary key; Access generates IDs
Title	Short Text	Book title
Author	Short Text	Author's name
Category	Short Text	For example, Programming or Networking

Field Name	Data Type	Notes
AvailableCopies	Number	Number of copies currently available

For a Number field, the default field size is usually Long Integer in many Access versions; this is suitable for AvailableCopies. You can leave the default unless your instructor specifies otherwise.

Screenshot 2: Capture Books in Design View, showing all field names, data types, and the key icon beside BookID.

3. Create the Members table

1. Choose **Create > Table Design** again.

2. Enter the fields below.

3. Select MemberID and click **Primary Key**.

4. Press Ctrl+S and save the table as **Members**.

Field Name	Data Type	Notes
MemberID	AutoNumber	Primary key
FullName	Short Text	Member's full name
Department	Short Text	For example, MCA or BCA
Phone	Short Text	Phone numbers should be text, not arithmetic values

Screenshot 3: Capture Members in Design View with its field names and primary key.

4. Create the BookIssues table

1. Choose **Create > Table Design**.

2. Add the fields shown below.

3. Select IssueID and click **Primary Key**.

4. For BookID and MemberID, select the Number data type. In the Field Properties pane, set **Field Size** to **Long Integer**. This is important because the referenced primary keys use AutoNumber.

5. ReturnDate can be left blank until the book is returned.

6. Press Ctrl+S and save the table as **BookIssues**.

Field Name	Data Type	Notes
IssueID	AutoNumber	Primary key
BookID	Number	Foreign key to Books.BookID; Field Size: Long Integer
MemberID	Number	Foreign key to Members.MemberID; Field Size: Long Integer
IssueDate	Date/Time	Date the book was borrowed
DueDate	Date/Time	Date the book should be returned
ReturnDate	Date/Time	Leave blank until returned

Screenshot 4: Capture BookIssues in Design View. Try to keep the field list and the Field Size setting visible if possible.

5. Enter sample records

Open a table by double-clicking it in the Navigation Pane. Enter data in Datasheet View. Do not type values into AutoNumber fields; Access creates those IDs automatically.

5.1 Books sample data

Title	Author	Category	AvailableCopies
Python Basics	John Smith	Programming	4
Computer Networks	Andrew Tanenbaum	Networking	2
Database Systems	Korth	Database	3
Linux Essentials	Jason Cannon	Linux	5

5.2 Members sample data

FullName	Department	Phone
Anu Thomas	MCA	9876500001
Rahul Nair	BCA	9876500002
Meera Joseph	MCA	9876500003

Use fictional sample data for this practice project. Your MemberID values will be assigned by Access; note the IDs shown in the table so you can use them in BookIssues.

5.3 BookIssues sample data

Enter two or three issue records using BookID and MemberID values that actually exist in your Books and Members tables. Example: if Python Basics has BookID 1 and Anu Thomas has MemberID 1, you can create an issue for those IDs. Your generated IDs may differ if you entered records in a different order.

BookID	MemberID	IssueDate	DueDate	ReturnDate
1	1	A recent date	A date 7 days later	(leave blank)
2	2	A recent date	A date 7 days later	(leave blank)
3	3	A date in the past	A date in the past	(leave blank)

The date descriptions above are instructions, not literal text to type. Enter actual dates using your Access date format. Use one past-due record to test the overdue query later.

Screenshots 5–7: Capture the Books, Members, and BookIssues tables in Datasheet View with sample rows visible. Avoid showing any real personal phone numbers.

6. Create relationships

1. Close any open table tabs.

2. Select **Database Tools > Relationships**.

3. If the Show Table dialog appears, add Books, Members, and BookIssues, then close the dialog.

4. Drag **Books.BookID** onto **BookIssues.BookID**.

5. In the Edit Relationships dialog, tick **Enforce Referential Integrity**, then click **Create**.

6. Drag **Members.MemberID** onto **BookIssues.MemberID**.

7. Tick **Enforce Referential Integrity**, then click **Create**.

8. Arrange the tables neatly so both relationships are visible, then press Ctrl+S to save the layout.

> You should see a 1 on the Books and Members sides, and an infinity symbol (∞) on the BookIssues side. Each book or member can appear in multiple issue records. If Access refuses to create a relationship, check that the foreign-key fields are Number with Field Size Long Integer, and that existing issue IDs match real Books and Members IDs.

Screenshot 8: Capture the Relationships window with all three tables and both 1-to-many relationships visible.

7. Create the queries

A query retrieves records that match criteria. For each query, use Create > Query Design, add the specified tables, add fields to the grid, enter the criteria, and save with the requested name. To test a query, click Run (the red exclamation mark) or switch to Datasheet View.

7.1 Query: Available Books

1. Choose **Create > Query Design** and add the Books table.

2. Double-click Title, Author, Category, and AvailableCopies to add them to the grid.

3. In the Criteria row under AvailableCopies, type **>0**.

4. Click Run. The query should show only books with more than zero available copies.

5. Save as **qryAvailableBooks**.

Screenshot 9: Capture the query results showing the available books.

7.2 Query: Issued Books

1. Choose Create > Query Design and add Books and BookIssues.

2. Access should join them using BookID if the relationship exists. If it does not, drag Books.BookID to BookIssues.BookID.

3. Add Title from Books and MemberID, IssueDate, and DueDate from BookIssues.

4. Add ReturnDate from BookIssues to the grid too. In its Criteria row, type **Is Null**. You may untick Show for ReturnDate if you do not want the field displayed.

5. Click Run and save as **qryIssuedBooks**.

This query finds issue records with no ReturnDate, meaning they are currently marked as not returned. For a more readable report later, you can also include FullName by adding Members and joining MemberID.

Screenshot 10: Capture the Issued Books query results.

7.3 Query: Overdue Books

1. Create another query using Books and BookIssues, with the same join on BookID.

2. Add Title, MemberID, IssueDate, DueDate, and ReturnDate.

3. Under DueDate, in the Criteria row, type **<Date()**.

4. Under ReturnDate, in the same Criteria row, type **Is Null**. Because both criteria are on the same row, both conditions must be true.

5. Click Run and save as **qryOverdueBooks**.

> If no rows appear, check that you entered at least one issue with a DueDate before today and left ReturnDate blank. Make sure the criteria are on the same row.

Screenshot 11: Capture the Overdue Books query results. It is okay if the result is empty only if there are genuinely no matching records, but for your demonstration, create one past-due sample record.

8. Create data-entry forms

Forms give users a friendly screen for entering or editing records without working directly in a table. Create one form per table using Access's Form command.

8.1 Books form

1. In the Navigation Pane, click the Books table once.

2. Choose **Create > Form**.

3. Access generates a form automatically. Check that Title, Author, Category, and AvailableCopies appear.

4. Press Ctrl+S and save as **frmBooks**.

5. Switch to Form View if needed. Use the record navigation buttons at the bottom to move between records, or use the New (blank) record button to add a test book.

Screenshot 12: Capture frmBooks in Form View with one sample book displayed. Keep the form title and its fields visible.

8.2 Members form

1. Select the Members table in the Navigation Pane.

2. Choose Create > Form.

3. Save the form as **frmMembers**.

4. Open it in Form View and check that FullName, Department, and Phone are visible.

Screenshot 13: Capture frmMembers in Form View with a sample member displayed.

8.3 BookIssues form

1. Select the BookIssues table and choose Create > Form.

2. Save the form as **frmBookIssues**.

3. Open the form and confirm that BookID, MemberID, IssueDate, DueDate, and ReturnDate appear.

4. Use existing BookID and MemberID values. Leave ReturnDate empty for a book that has not been returned.

> Beginner note: the automatically generated form may show numeric IDs instead of book titles and member names. That is acceptable for the basic version. A Combo Box lookup is an optional enhancement, not required to complete this guide.

Screenshot 14: Capture frmBookIssues in Form View. Ideally show a sample issue record with BookID, MemberID, IssueDate, and DueDate.

9. Create a printable report

1. In the Navigation Pane, select **qryIssuedBooks**.

2. Choose **Create > Report Wizard**.

3. Move the fields you want from Available Fields to Selected Fields. Include Title, MemberID, IssueDate, and DueDate.

4. Choose a simple layout such as Tabular and click Finish.

5. Save the report as **rptIssuedBooks**.

6. Open Print Preview or Report View and check that the columns are readable.

Optional improvement: add Members to the query and include FullName instead of only MemberID. This makes the report easier to understand, but the basic report works with MemberID.

Screenshot 15: Capture rptIssuedBooks in Print Preview or Report View with the report heading and rows visible.

10. Final testing

Run through this checklist before submitting. If a test fails, fix it and capture the corrected result.

- ■ The database file is saved as LibraryManagement.accdb.
- ■ Books, Members, and BookIssues tables open successfully.
- ■ BookID, MemberID, and IssueID are primary keys.
- ■ Both relationships show one-to-many links and enforce referential integrity.
- ■ The Available Books query excludes books with zero copies.
- ■ The Issued Books query only shows records where ReturnDate is blank.
- ■ The Overdue Books query only shows records with DueDate before today and ReturnDate blank.
- ■ All three forms open and show the expected fields.
- ■ The report opens and displays issue records.
- ■ The database is saved and closed cleanly.

11. Screenshot and submission guide

Your instructor likely wants evidence that you actually created and used the database interface. Do not submit only screenshots of the final result. Capture the important stages as you build the project.

Screenshot	What must be visible
1	Database opened in Microsoft Access; project name visible
2–4	Each table's Design View, including fields and primary key
5–7	Sample data in Datasheet View for all three tables
8	Relationships window showing both links
9–11	Each query's results in Datasheet View
12–14	Each form in Form View, with fields and a sample record
15	Report in Print Preview or Report View

• On Windows, press **Win + Shift + S** to open Snipping Tool. Drag around the relevant Access window, then save the capture as a PNG.

• Create a folder named **AccessProjectScreenshots** and use descriptive names such as **01-database.png**, **02-books-design.png**, and **12-books-form.png**.

• Make sure text is readable, the relevant Access ribbon or object name is visible, and no unrelated windows cover the interface.

• Take genuine screenshots from your own work. Do not fabricate or edit screenshots to imply a feature works when it does not.

• If your sir wants a written record, place the screenshots in order and add a short caption below each one describing what the screenshot demonstrates.

12. Common problems and fixes

Problem	Likely cause and fix
Cannot create a relationship	Check that the foreign key is Number with Field Size Long Integer, and that every non-empty foreign-key value exists in the parent table.
Query returns no rows	Check your sample data and criteria. For overdue books, ensure DueDate is before today and ReturnDate is blank.
Access asks for a parameter	A field name may be misspelled or the field may not have been added. Check spelling and table names.
Form does not show new records	Save the table and form, then move to the last/new record or close and reopen the form.
Report is blank	Open the query the report uses and confirm it returns rows. Then reopen the report.
IDs do not start at 1	AutoNumber values can have gaps. Use the actual IDs displayed in your tables rather than assuming their values.

13. What to say during a demonstration

Use this short explanation in your own words:

> “My project is a Library Management System created in Microsoft Access. The Books table stores book details, the Members table stores library member details, and the BookIssues table records borrowing transactions. Primary keys uniquely identify records, while relationships connect the issue records to the relevant books and members. Queries filter available, issued, and overdue books. Forms provide a user-friendly way to enter records, and the report presents issued-book information in a printable format.”

Be honest about the scope: this beginner version records issue and return dates, but it does not automatically adjust AvailableCopies or prevent duplicate active loans. Those would be future enhancements.

14. File naming and final save

- Main database: **LibraryManagement.accdb**
- Tables: **Books**, **Members**, **BookIssues**
- Queries: **qryAvailableBooks**, **qryIssuedBooks**, **qryOverdueBooks**
- Forms: **frmBooks**, **frmMembers**, **frmBookIssues**
- Report: **rptIssuedBooks**
- Screenshots folder: **AccessProjectScreenshots**

> Press Ctrl+S after editing objects. Close open forms, tables, and reports before copying the .accdb file for submission. Keep a backup copy.

End of guide